

The Grand Illusion

Why Quantum Entanglement Demands a New Architecture

for Space and Time

Spin Fabric Theory

Rick G. Braddy

Independent Researcher

May 2026

Abstract

We propose that there exists a real physical medium — a structured vacuum fabric — from which space, time, and matter emerge. In this framework, gravity is a collective fabric distortion, entanglement is a shared structural connection within the fabric, and time is a relational property arising from quantum correlations between subsystems. This organizing hypothesis synthesizes three independently peer-reviewed programs: Einstein–Cartan torsion theory (intrinsic spin as a geometric source), holographic entanglement entropy (spacetime connectivity as entanglement structure), and the Page–Wootters mechanism (time from quantum entanglement, experimentally confirmed 2014). We demonstrate that quantum entanglement — confirmed across every physical medium and across fourteen orders of magnitude in distance — demands a reframing of both space and time as emergent rather than fundamental. We extend the framework to black holes, identifying them as maximum fabric condensation events whose structure is explained by the same principles. One near-term falsifying experiment is identified. The framework is required to reduce to QED and general relativity in their established domains and makes no claims about Standard Model substructure, antimatter, or the dark sector.

1. The Core Model: Principles of the Spin Fabric

We begin with the organizing principles of the Spin Fabric framework stated plainly. Each is either supported by established peer-reviewed physics or is a natural extension of it. The principles are ordered from most fundamental to most derived.

1.1 The Fabric is Fundamental

What we call “empty space” is not empty. The quantum vacuum carries measurable structure: the Casimir effect [1, 2] demonstrates that vacuum mode restriction between conducting boundaries produces measurable forces, and the ATLAS observation of light-by-light scattering at the LHC [3] confirms that the vacuum is nonlinear, exactly as QED predicts. The fabric is real, structured, and responsive to disturbances.

1.2 c is the Fabric’s Maximum Propagation Speed

There is a maximum natural propagation speed inherent to the fabric itself. We call it c — the speed of light — because light was the first thing measured traveling at that speed. It is more precisely understood as the maximum signal speed of the medium: the speed at which massless disturbances propagate through the fabric. This is consistent with its derivation from Maxwell's equations and its role as the universal speed limit in special relativity.

1.3 Massless and Massive Modes

Energy manifests in the fabric in two fundamental modes. Massless energy (photons, electromagnetic radiation) consists of pure propagating disturbances traveling through the fabric at speed c . They carry no rest mass and, as QED describes, travel along null geodesics whose proper time is zero. **Massive energy** (electrons, protons, atoms, planets) consists of localized, stable configurations that interact strongly with the fabric, producing inertia and preventing acceleration to c . $E=mc^2$ expresses the equivalence between these two modes: matter-antimatter annihilation — a measured, reproducible phenomenon — demonstrates that trapped massive configurations can convert entirely to free propagating energy.

1.4 Time is Emergent

Time is not a universal backdrop through which events occur. Page and Wootters (1983) [4] showed theoretically that temporal ordering arises from quantum entanglement between subsystems rather than from an external clock. Moreva et al. (2014) [5] confirmed this experimentally: within a closed quantum system, an internal observer (one photon acting as a clock) perceives time evolving through its entanglement with the other photon, while an external observer sees a stationary, timeless state. Time, in this framework, is what entanglement looks like from the inside. Gravitational time dilation — the measurable slowing of clocks in stronger gravitational fields, confirmed by GPS satellites — reflects the curvature of local spacetime geometry, consistent with general relativity and naturally expressed as the fabric's geometric response to mass-energy density.

1.5 Space and Distance are Emergent

What we perceive as spatial separation is a derived property of how the fabric is configured. Van Raamsdonk (2010) [10] showed that spacetime connectivity in the holographic correspondence is supported by entanglement: progressively disentangling a quantum system causes the corresponding spacetime geometry to disconnect. Distance, in this framework, is not the fundamental layer — entanglement structure is.

1.6 Entanglement is Structural Connection

When two particles become entangled, they create a shared, coherent structure in the fabric. The correlation is not transmitted across space — it exists as a direct structural connection within the fabric itself. This is why the effect appears instantaneous: at the fabric level, the two particles were never fully separated. The ER=EPR conjecture [11] gives this a geometric form: entangled

pairs are connected by non-traversable Einstein–Rosen bridges. Non-traversable means no signaling; real means the correlation has a structural origin. This is consistent with all Bell test results and the no-signaling theorem.

1.7 Gravity is Collective Fabric Distortion

Large masses create a broad, collective distortion in the fabric through their total energy density. What we feel as gravity is the fabric’s geometric response to that accumulated distortion — precisely as general relativity describes through spacetime curvature. Einstein–Cartan theory [13] extends this minimally by incorporating intrinsic spin: the canonical spin density tensor sources algebraic torsion, adding a spin-coherence channel to the geometric response that vanishes for ordinary unpolarized matter and recovers GR exactly.

1.8 The Core Philosophical Point

We are embedded inside the fabric. Our perception of space, time, and causality are all emergent phenomena arising from our position within the medium. This is why nonlocal effects like entanglement feel “spooky” — they violate the rules of the emergent layer we inhabit, not the rules of the fundamental layer. The Grand Illusion is that the emergent layer is fundamental.

Figure 1. Convergence of three empirical anchors on the central hypothesis

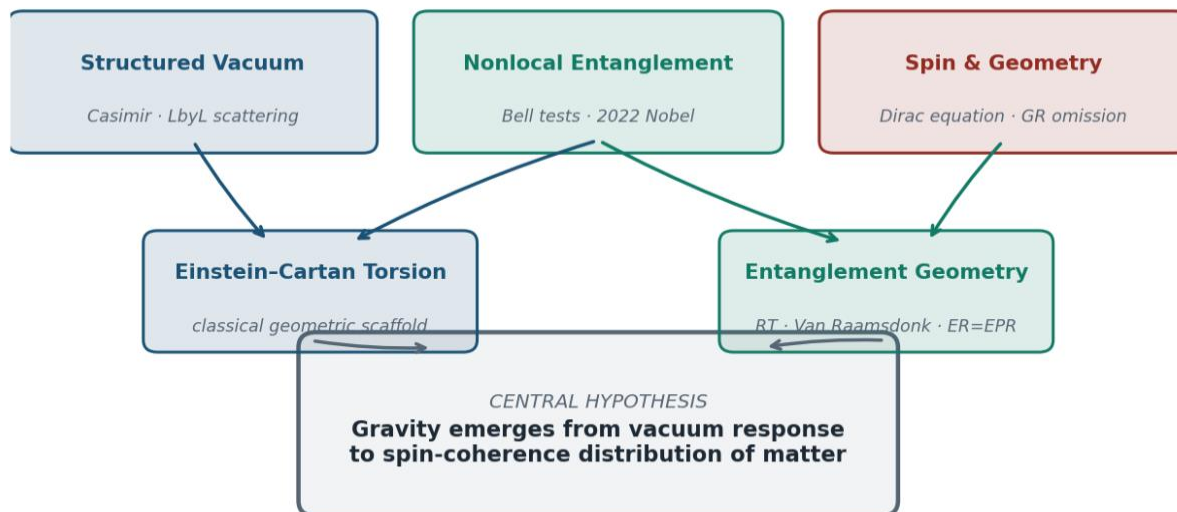


Figure 1. The Spin Fabric framework connects three independently peer-reviewed programs — Einstein–Cartan torsion, holographic entanglement entropy, and the Page–Wootters mechanism — through spin coherence as the matter-side meeting point.

2. The Evidence: Entanglement Across Every Medium and Scale

The organizing premise of the Spin Fabric framework — that the fabric is real, that entanglement is structural, and that time is emergent — rests on experimental evidence that is among the most precisely confirmed in all of physics. Entanglement has been demonstrated across every physical medium accessible to experiment, across fourteen orders of magnitude in distance, across a temperature range from millikelvin to room temperature, and in systems ranging from single photons to objects composed of trillions of atoms. This universality is not incidental. It is the argument.

A medium-specific artifact would appear in one substrate and not others. A temperature-dependent classical correlation would disappear at room temperature. A size-limited effect would break down for macroscopic objects. Entanglement does none of these things. The same nonlocal correlations — the same violation of Bell's inequalities by the same quantum-mechanical prediction — have been observed across systems so different in every physical property that the only common factor is the correlations themselves. That is what universality means, and it demands a universal explanation.

2.1 Distance: From Micrometers to Megameters

Ion traps ($\sim 10^{-6}$ m). Rowe et al. (2001) [9] demonstrated entanglement between beryllium ions in a trap, establishing the phenomenon for massive charged particles at the scale of individual atomic positions. The entanglement separation here is measured in micrometers — the short end of our distance range.

Laboratory to city scale (1 m – 1.3 km). Aspect, Dalibard, and Roger (1982) [6] performed the first time-varying Bell test with entangled photons, closing the locality loophole with detectors 13 meters apart and switching analyzers faster than light could travel between them. Hensen et al. (2015) [12] extended this to 1.3 km using electron spins in nitrogen-vacancy centers in diamond, simultaneously closing both the locality and detection loopholes in a single experiment — the first fully loophole-free Bell test. The 2022 Nobel Prize recognized Aspect, Clauser, and Zeilinger for this collective program.

City to continental scale (143 km – 1,400 km). Zeilinger's group demonstrated free-space entanglement over 143 km between the Canary Islands (Ma et al. 2012 [7]). The Micius satellite program then achieved entanglement distribution at 1,200 km of free space (Yin et al. 2017 [8]) and ground-to-satellite uplinks exceeding 1,400 km. At these distances, any light-speed signal explanation of the correlation would require the particles to have communicated their measurement settings before they were made. The correlations arrive faster than that.

2.2 Medium: Every Physical Substrate

Photons in vacuum and fiber. The canonical medium for Bell tests. Entanglement has been demonstrated in optical fiber, free space, and through the atmosphere, with the photon pairs generated by spontaneous parametric down-conversion in nonlinear crystals. Photons are bosons; their entanglement involves polarization, a spin-like degree of freedom.

Trapped ions (massive fermions). Rowe et al. [9] demonstrated that entanglement applies equally to massive charged particles in controlled quantum states — not a quirk of massless light.

Electron spins in solid-state matter. Hensen et al. [12] demonstrated entanglement between electron spins bound to nitrogen-vacancy defects in solid diamond at room temperature, 1.3 km apart. The spin is the entangled degree of freedom — directly relevant to the Spin Fabric framework's emphasis on intrinsic spin as a geometric source.

Superconducting qubits (collective quantum states). Ansmann et al. (2009) [14] demonstrated Bell inequality violation in superconducting Josephson junction circuits at millikelvin temperatures. The entangled entities here are not individual particles but macroscopic collective quantum states — Cooper pairs of electrons forming a single quantum object. This extends entanglement from the particle level to the collective condensate level.

Atomic ensembles. Matsukevich et al. (2006) [21] demonstrated entanglement between atomic ensembles — large collections of atoms — not individual particles. This shows that entanglement applies to bulk matter at the ensemble level, not merely to individually isolated quantum systems.

Phonons in macroscopic diamonds at room temperature. Lee et al. (2011) [22] reported strong nonclassical correlations between the vibrational modes (phonons) of two millimeter-sized diamond crystals at room temperature, separated by 15 cm. The quantum state of the diamonds was shown to have positive concurrence with 98% probability. This is among the most striking demonstrations of entanglement-class correlations: ordinary diamonds, at 20°C, exhibiting behavior whose explanation requires quantum correlations between their atomic vibrations. The experiment has been subject to interpretation debate regarding whether the correlations constitute true nonlocal entanglement or path-indistinguishability effects; the nonclassical correlations themselves are not disputed.

Mechanical oscillators ($\sim 10^{12}$ atoms). Ockeloen-Korppi et al. (2018) [23] demonstrated stabilized entanglement between two massive micromechanical oscillators — each composed of approximately 10^{12} atoms — coupled to a microwave electromagnetic cavity. The Duan criterion for entanglement was violated by 1.4 dB. This is the largest-mass system in which entanglement has been quantifiably confirmed; the entangled entities are macroscopic solid objects, not elementary particles.

2.3 Temperature: From Millikelvin to Room Temperature

A classical thermal effect would be suppressed or eliminated as temperature increases. The entanglement evidence spans the full available temperature range. Superconducting qubit experiments operate at 15 millikelvin [14]. NV-center spin experiments operate at room temperature [12]. Diamond phonon experiments operate at room temperature [22]. Mechanical oscillator experiments operate at millikelvin temperatures but entangle objects of macroscopic mass [23]. The correlations persist across this range. Temperature does not eliminate them; it only changes the experimental difficulty of demonstrating them.

2.4 Time: The Page–Wootters Experiment

The most profound extension. Moreva et al. (2014) [5] experimentally confirmed the Page–Wootters (1983) [4] mechanism: temporal ordering itself emerges from quantum entanglement between subsystems. A photon pair was treated as a closed quantum system. One photon served as an internal clock; the other as the observed system. From the perspective of the internal clock photon, the other photon evolved in time. From the perspective of an external observer not entangled with the system, the entire system was stationary — no time evolution at all. Time, in this experiment, is not a background against which quantum systems evolve. It is a relational property arising from entanglement.

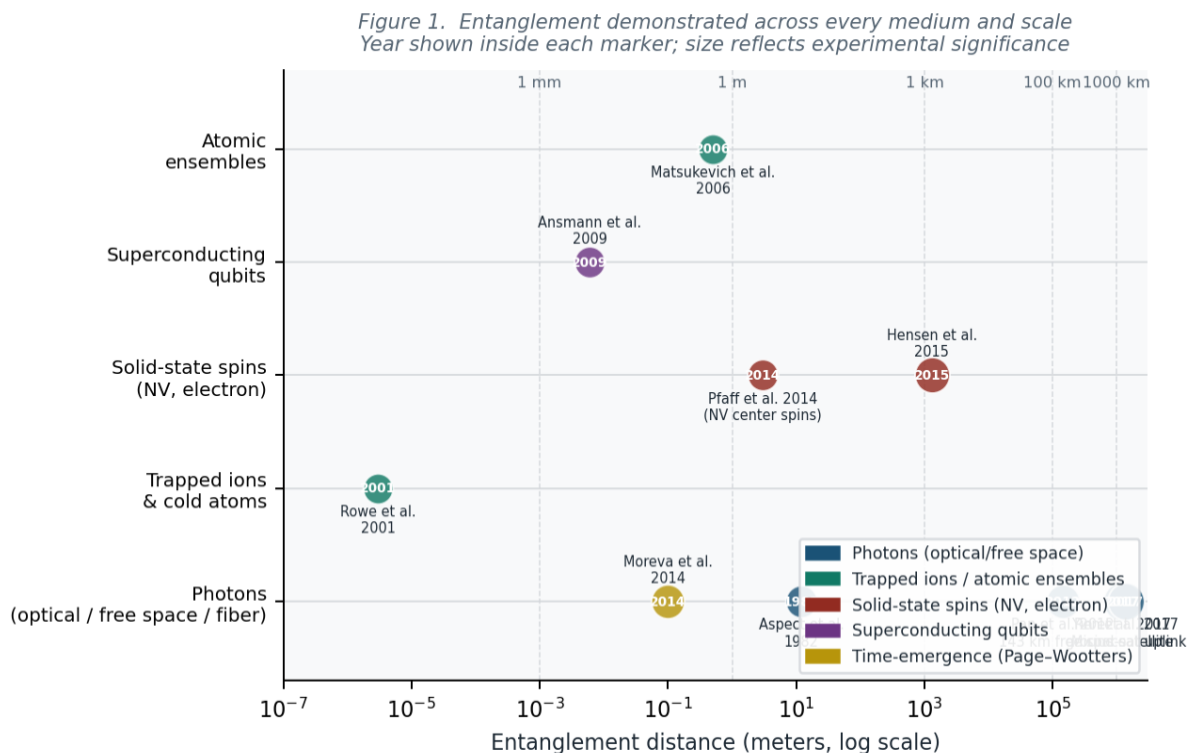


Figure 2. Entanglement demonstrated across every physical medium and across fourteen orders of magnitude in distance. Each data point represents a distinct experimental substrate, temperature regime, and particle type. Moreva et al. (gold) is distinguished as experimental confirmation that time itself is entanglement-emergent.

2.5 Why Universality Is the Argument

Consider what would be required for the entanglement correlations to have a medium-specific classical explanation. A mechanism confined to photons would not explain ion trap entanglement. A mechanism that required cryogenic temperatures would not explain room-temperature diamond correlations. A mechanism limited to microscopic particles would not explain 10^{12} -atom mechanical oscillator entanglement. A mechanism operating only in space would not explain the Page–Wootters temporal result.

No single classical mechanism accounts for all of these simultaneously. The only unified account is the quantum-mechanical one: these systems share correlations because they share a quantum state that cannot be factored into independent local descriptions. In the Spin Fabric framework, this is equivalent to saying they share a structural connection in the fabric — a connection that exists at the fundamental layer, prior to the emergent spatial and temporal separation we observe from within that layer.

Bell's theorem [24] proved that no local hidden-variable theory reproduces the quantum predictions. Loophole-free experiments [12] confirmed the quantum predictions. The 2022 Nobel Prize formalized the conclusion. The universe exhibits nonlocal correlations that no local realistic theory can explain. This is not a gap in our understanding. It is a door into a deeper one.

3. What the Evidence Demands of Time

The standard response to Bell's theorem is: "the correlations are real but no information is transmitted, so special relativity is safe." This is correct as far as it goes. But it is not an adequate stopping point. It does not explain what entanglement is. It merely describes what it cannot do.

The Page–Wootters result goes further. Time, in that experiment, is not a background parameter. It is a relational property: the ordering of events arises from the entanglement between subsystems. An external observer, not entangled with the system, sees no time evolution at all. Time is what entanglement looks like from the inside.

Van Raamsdonk (2010) [10] showed that spacetime connectivity is supported by entanglement. Jacobson (1995) [15] derived the Einstein field equations from the thermodynamics of local horizons, treating gravity as an equation of state. Ryu and Takayanagi (2006) [16] established that entanglement entropy equals geometric area in the holographic limit. These results collectively demand a specific conclusion: time and space are not fundamental containers. They are emergent structures arising from a deeper layer whose primary ontology is entanglement.

The reframing. Two entangled particles do not influence one another across space and time. They were never fully separated in the fabric. The distance between them is an emergent property — real and measurable at the level of classical physics, but not the fundamental description. At the fabric level, they share a single coherent structure. When that structure is disturbed at one end, the state at the other end is not updated — it was always part of the same thing.

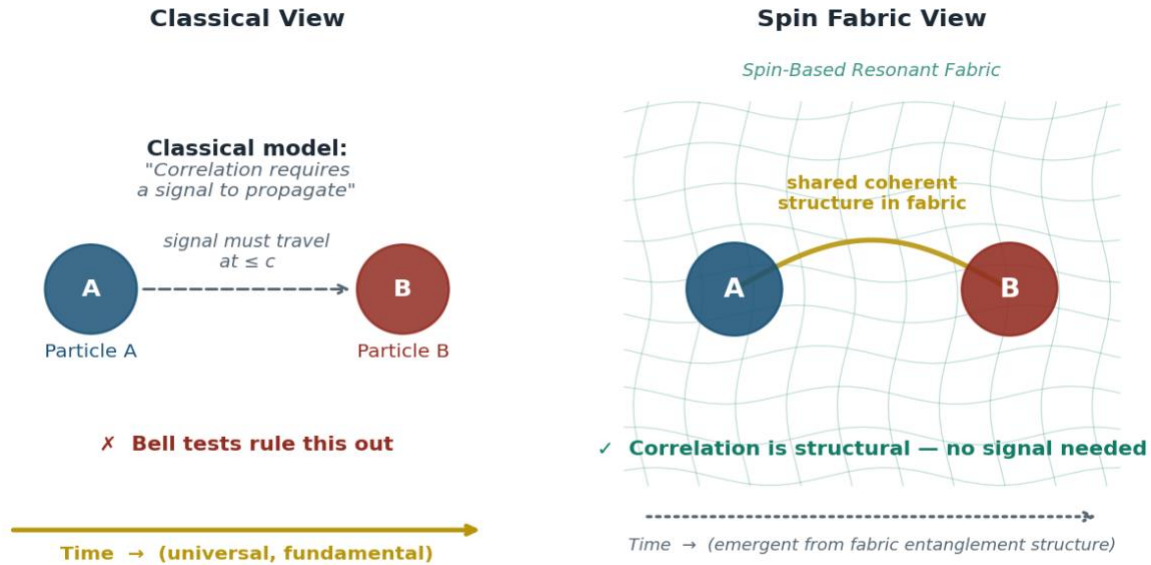


Figure 2. Classical physics requires a propagating signal to explain correlations. The spin fabric framework explains them as structural: both particles share the same fabric excitation; time itself is emergent from that structure.

Figure 3. Classical physics requires a propagating signal to explain correlations between distant particles (left). The Spin Fabric framework explains them as structural: both particles share the same fabric excitation; time and distance are emergent from the fabric, not the fundamental layer (right).

4. The Framework: Three Peer-Reviewed Pillars

The Spin Fabric hypothesis connects three independently established programs in physics. It does not invent new physics. It proposes that these three frontiers describe facets of a single underlying picture.

4.1 Einstein–Cartan Torsion Theory

Einstein–Cartan (EC) theory [13] is the minimal classical extension of general relativity that incorporates intrinsic spin. The torsion tensor is sourced algebraically by the canonical spin density tensor of matter fields. For ordinary unpolarized matter — paired electron spins, random nuclear orientations — the spin density tensor vanishes on expectation, torsion vanishes, and EC reduces exactly to GR. The relevant source quantity is spin coherence: a quantum-information property of the matter state, not spin density alone. At extreme densities — neutron star interiors, the very early universe, engineered coherent spin systems — the torsion contribution becomes non-negligible. Poplawski’s cosmological work [17] shows EC torsion averts the initial singularity through spin-induced repulsion.

4.2 Holographic Entanglement Entropy

Ryu and Takayanagi (2006) [16] showed that in the AdS/CFT correspondence, the entanglement entropy of a boundary region equals the area of a minimal surface in the bulk spacetime. Entanglement entropy and geometric area are literally equal. Van Raamsdonk (2010) [10] showed that reducing entanglement in a holographic CFT disconnects the corresponding bulk geometry.

Maldacena and Susskind (2013) [11] proposed ER=EPR: entangled pairs are connected by non-traversable Einstein–Rosen bridges. Spacetime geometry is entanglement structure.

4.3 The Page–Wootters Mechanism

Page and Wootters (1983) [4] showed that time evolution can be described entirely through correlations between subsystems of a closed quantum system, without reference to an external clock. Moreva et al. (2014) [5] confirmed this experimentally. This is the cleanest peer-reviewed demonstration that time is emergent from quantum correlations — from entanglement in the fabric.

4.4 The Meeting Point: Spin Coherence

Both EC theory and the entanglement-geometry program independently identify the quantum coherence of the matter state as relevant to the local geometric response. EC arrives there through the spin density tensor; the entanglement-geometry program arrives there through the dependence of entanglement structure on the matter density matrix. The Spin Fabric hypothesis is that this convergence is not coincidental — both describe the same physics from different angles.

5. Black Holes in the Spin Fabric

Black holes are the most extreme fabric configurations in the observable universe. Understanding them within the Spin Fabric framework both tests the framework’s consistency and reveals its most distinctive predictions.

5.1 Maximum Fabric Condensation

In the Spin Fabric picture, a black hole is a region where the fabric’s distortion reaches its maximum — the point where no free spin wave (massless excitation) can propagate outward. The event horizon is the boundary where the fabric’s distortion speed exceeds c : all inward-propagating excitations are captured; none escape. Within the horizon, all infalling energy — both free propagating waves (photons) and localized massive configurations (matter) — becomes part of the maximum-density trapped structure. Infalling photons contribute their energy to the black hole’s total mass. In this sense, black holes are extreme converters of free propagating energy into maximally trapped configurations.

5.2 No Singularity: The EC Torsion Core

In pure general relativity, the singularity — a point of infinite density — is unavoidable inside a black hole. Einstein–Cartan theory changes this. As density increases toward extreme values, the spin density of compressed matter becomes non-negligible, generating a torsion-induced repulsive term in the effective stress-energy. Poplawski [17] has shown that this EC repulsion averts the singularity: there is a maximum density, not an infinite one. Inside the event horizon, at

maximum spin-coherence densities, the EC torsion term reaches its global maximum — making black holes the most extreme source of the spin-geometry coupling the framework predicts.

5.3 Information: Preserved in the Fabric

The information paradox asks: what happens to the quantum information describing everything that falls into a black hole? Standard GR offers no resolution. The Spin Fabric framework, via ER=EPR [11], provides one. Hawking pairs — particle-antiparticle pairs created at the event horizon, one falling in, one escaping as Hawking radiation — are entangled. In the fabric picture, this entanglement is a fabric connection: a micro-ER bridge linking the interior to the escaping radiation. Information is not lost; it is encoded in the entanglement structure of the fabric — in the topology of connections between the black hole's interior and its radiation field. This is consistent with the holographic principle: the information content of a black hole is proportional to its horizon area, and that area is an entanglement entropy (Ryu–Takayanagi).

5.4 Hawking Radiation as Fabric Unraveling

Hawking radiation, in the Spin Fabric picture, is the slow quantum unraveling of a maximum trapped configuration back into free propagating spin waves. Over cosmic timescales, black holes are therefore net converters of trapped massive energy back toward free propagating energy — the reverse of their accretion phase. The full lifecycle of a black hole runs both directions: free-to-trapped during accretion, trapped-to-free during evaporation. This is $E=mc^2$ operating at cosmic scales and timescales, mediated by the fabric's quantum geometry.

6. One Testable Prediction

The framework produces a specific prediction distinguishable from generic quantum gravity. The active Quantum Gravity-induced Entanglement of Masses (QGEM/BMV) program [18, 19] places two massive nanoparticles in spatial superposition via Stern–Gerlach forces — which are spin-mediated — and looks for gravitational entanglement between them as a witness of quantum gravity.

Generic quantum gravity predicts that the entanglement generation rate depends on mass and separation, independent of whether the superposition mechanism is spin-mediated. The Spin Fabric framework predicts that spin-mediated superpositions generate entanglement at a higher rate per unit mass than equivalent non-spin-mediated superpositions (e.g., matter-grating diffraction of neutral fullerenes), because spin coherence directly amplifies the geometric coupling through the EC torsion sector.

The distinguishing experiment. Run the QGEM experiment in two parallel configurations — NV-center diamond (spin-mediated) and neutral matter-grating (non-spin-mediated) — and compare entanglement generation rates per unit mass. If $R_{\text{spin}} > R_{\text{grating}}$, the framework is supported. If $R_{\text{spin}} = R_{\text{grating}}$, it is falsified. The same consortium has the capability to run both variants [20].

A second channel: Bell-test correlations for entangled photon pairs measured near a magnetically saturated sample should differ from the same measurement near a demagnetized control of identical mass-energy. The local entanglement structure of the fabric, modified by the coherent spin density, should shift the Bell parameter. This is accessible in principle with existing entangled-photon apparatus and a controllable magnetic environment.

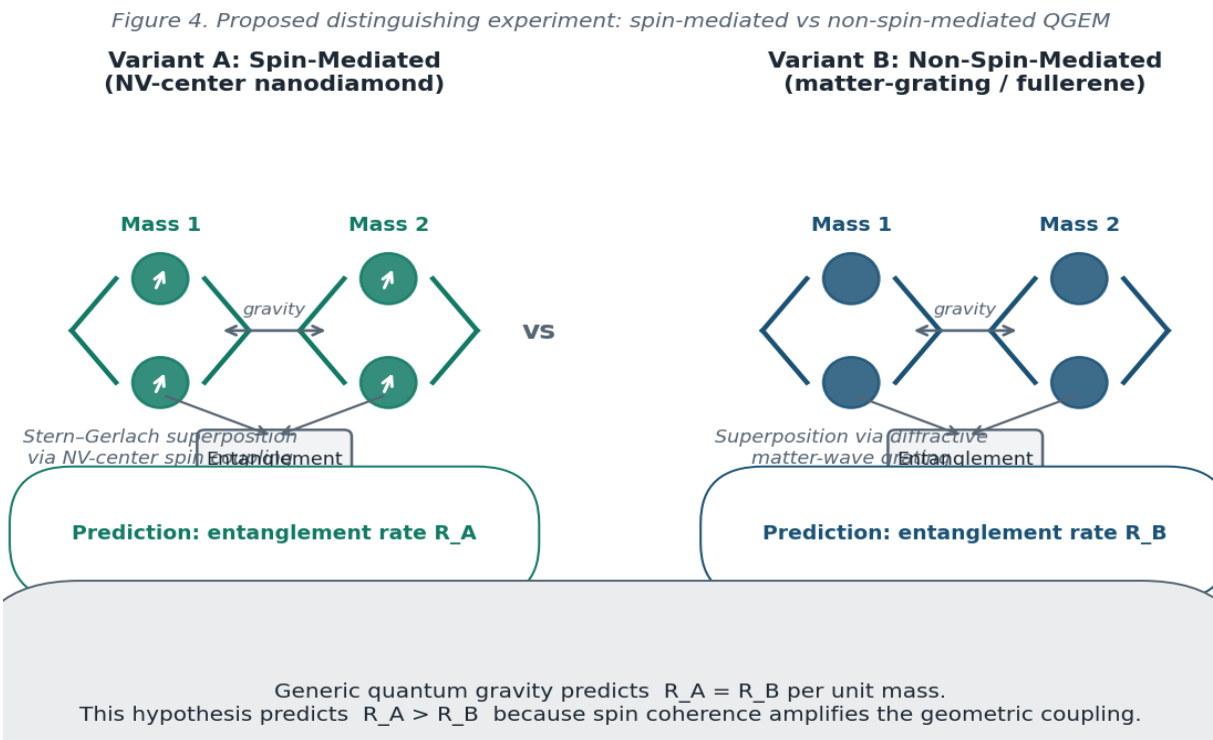


Figure 4. The proposed distinguishing experiment. Variant A uses NV-center spin to create superposition (spin-mediated). Variant B uses matter-grating diffraction (non-spin-mediated). Generic quantum gravity predicts $R_{spin} = R_{grater}$. The Spin Fabric framework predicts $R_{spin} > R_{grater}$.

Figure 3. Sensitivity landscape: predicted EC torsion signal vs current and near-term experimental reach

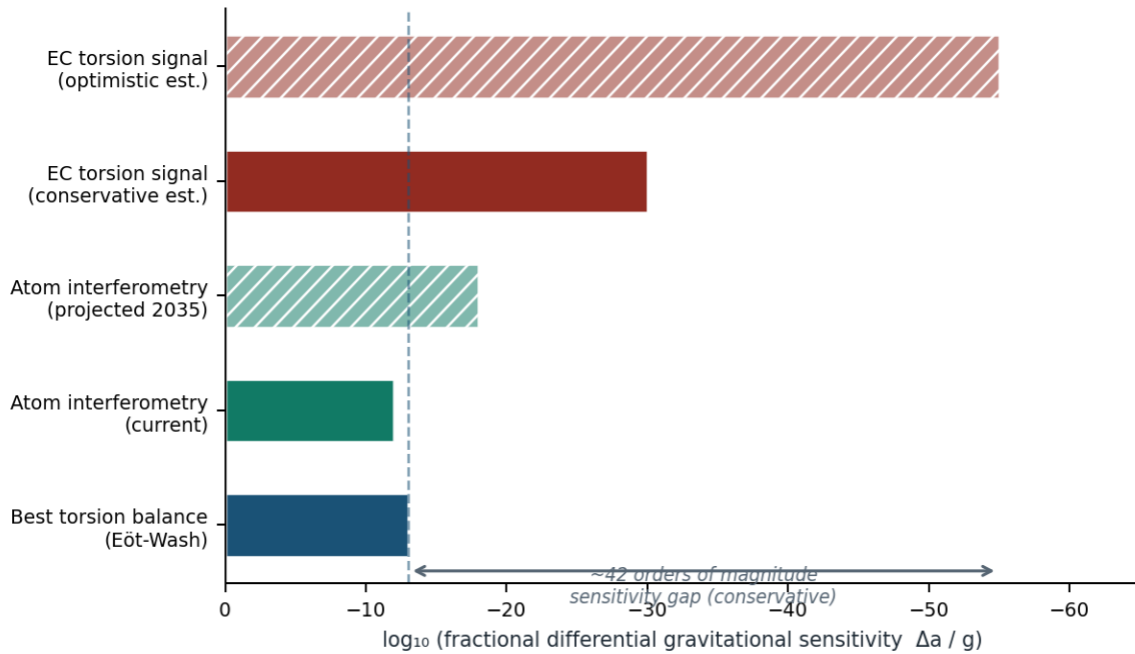


Figure 5. Experimental sensitivity landscape. Current gravitational measurement reach (upper bars) is many orders of magnitude above the predicted EC torsion signal (lower bars). The productive near-term falsification path runs through quantum-information channels and the QGEM distinguishing experiment, not static gravimetry.

7. What the Spin Fabric Framework Is Not

To prevent the failure modes endemic to speculative physics, we state explicitly what this framework does not claim.

- **Not a replacement for the Standard Model.** No new substructures beneath quarks and leptons. No elimination of antimatter. No claim about the dark sector.
- **Not a derivation of QED.** The framework is an organizing hypothesis, not a first-principles mathematical derivation. It requires reduction to QED in established domains.
- **Not currently falsifiable by static gravimetry.** The predicted spin-coherence gravitational signal is orders of magnitude below current static-field sensitivity. Falsification requires quantum-information channels and the QGEM distinguishing experiment.
- **Not a complete theory.** It is a hypothesis proposing that EC torsion, holographic entanglement entropy, and the Page–Wootters mechanism describe facets of one physics. The derivation connecting them is left to future work.

8. Conclusion

Spooky action at a distance is not an error in a local universe. It is the signature of the unified substrate from which our universe emerges. Space is not empty. Time is not universal. We are

localized disturbances propagating through a fabric whose entanglement structure is the geometry we inhabit.

The Spin Fabric framework proposes that gravity, entanglement, and the emergence of time are three expressions of a single underlying picture — one in which the quantum vacuum is a structured, long-range-entangled medium, spin coherence is the matter-side coupling to its geometric response, and the apparent locality of classical physics is a property of the emergent layer, not the fundamental one. Black holes, in this picture, are the fabric's most extreme configurations: maximum condensation events where the spin-geometry coupling reaches its global maximum, where EC torsion averts the singularity, and where Hawking evaporation is the slow return of trapped energy to free propagating waves.

The framework does not invent new physics. It is a structural bridge connecting three established frontiers. It produces one near-term falsifying experiment — the QGEM spin-mediated versus non-spin-mediated comparison — and a second channel in Bell-test measurements near coherent spin sources. We invite criticism, refinement, and falsification.

Acknowledgments

The author acknowledges that early exposure to independently developed speculative frameworks — including the Uniphics manuscript of P. J. Maley (2025) — helped identify intrinsic spin as a central organizing theme. The present framework is independently developed and anchored exclusively on the peer-reviewed literature cited herein; none of the specific particle-physics, substructure, or theory-of-everything commitments of that framework are adopted here.

Special thanks to contributions by Grok and Claude AI, along with NotebookLM for graphics.

References

- [1] Casimir, H. B. G. (1948). On the attraction between two perfectly conducting plates. *Proc. K. Ned. Akad. Wet.* 51, 793–795.
- [2] Lamoreaux, S. K. (1997). Demonstration of the Casimir force in the 0.6 to 6 μm range. *Phys. Rev. Lett.* 78, 5–8.
- [3] Aaboud, M., et al. (ATLAS Collaboration). (2017). Evidence for light-by-light scattering in heavy-ion collisions at the LHC. *Nature Physics* 13, 852–858.
- [4] Page, D. N., & Wootters, W. K. (1983). Evolution without evolution: Dynamics described by stationary observables. *Phys. Rev. D* 27, 2885–2892.
- [5] Moreva, E., et al. (2014). Time from quantum entanglement: An experimental illustration. *Phys. Rev. A* 89, 052122.
- [6] Aspect, A., Dalibard, J., & Roger, G. (1982). Experimental test of Bell's inequalities using time-varying analyzers. *Phys. Rev. Lett.* 49, 1804–1807.
- [7] Ma, X.-S., et al. (2012). Quantum teleportation over 143 kilometres using active feed-forward. *Nature* 489, 269–273.

- [8] Yin, J., et al. (2017). Satellite-based entanglement distribution over 1200 kilometres. *Science* 356, 1140–1144.
- [9] Rowe, M. A., et al. (2001). Experimental violation of a Bell's inequality with efficient detection. *Nature* 409, 791–794.
- [10] Van Raamsdonk, M. (2010). Building up spacetime with quantum entanglement. *Gen. Rel. Grav.* 42, 2323–2329.
- [11] Maldacena, J., & Susskind, L. (2013). Cool horizons for entangled black holes. *Fortschritte der Physik* 61, 781–811.
- [12] Hensen, B., et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature* 526, 682–686.
- [13] Hehl, F. W., von der Heyde, P., Kerlick, G. D., & Nester, J. M. (1976). General relativity with spin and torsion. *Rev. Mod. Phys.* 48, 393–416.
- [14] Ansmann, M., et al. (2009). Violation of Bell's inequality in Josephson phase qubits. *Nature* 461, 504–506.
- [15] Jacobson, T. (1995). Thermodynamics of spacetime: The Einstein equation of state. *Phys. Rev. Lett.* 75, 1260–1263.
- [16] Ryu, S., & Takayanagi, T. (2006). Holographic derivation of entanglement entropy from AdS/CFT. *Phys. Rev. Lett.* 96, 181602.
- [17] Poplawski, N. J. (2010). Cosmology with torsion: An alternative to cosmic inflation. *Phys. Lett. B* 694, 181–185.
- [18] Bose, S., et al. (2017). Spin entanglement witness for quantum gravity. *Phys. Rev. Lett.* 119, 240401.
- [19] Marletto, C., & Vedral, V. (2017). Gravitationally induced entanglement between two massive particles. *Phys. Rev. Lett.* 119, 240402.
- [20] Bose, S., et al. (2025). A spin-based pathway to testing the quantum nature of gravity. CERN ESPP White Paper.
- [21] Matsukevich, D. N., et al. (2006). Entanglement of remote atomic qubits. *Phys. Rev. Lett.* 96, 030405.
- [22] Lee, K. C., et al. (2011). Entangling macroscopic diamonds at room temperature. *Science* 334, 1253–1256. DOI: 10.1126/science.1211914.
- [23] Ockeloen-Korppi, C. F., et al. (2018). Stabilized entanglement of massive mechanical oscillators. *Nature* 556, 478–482. DOI: 10.1038/s41586-018-0038-x.
- [24] Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. *Physics* 1, 195–200.